

Memory and Long-Context Mechanisms for Long-Horizon LLM Agents

Abstract

The deployment of Large Language Model (LLM) agents in long-horizon tasks—ranging from complex software engineering and scientific discovery to lifelong personalized dialogue—has exposed fundamental limitations in standard autoregressive architectures. While context windows have expanded, the quadratic complexity of attention mechanisms, the degradation of retrieval precision in massive contexts, and the “lost in the middle” phenomenon necessitate sophisticated memory mechanisms beyond simple concatenation. This survey provides a comprehensive review of recent advancements in memory and long-context management for LLM agents, synthesizing evidence from 300 distinct academic studies. We propose a unified taxonomy categorizing approaches into hierarchical memory architectures, active context curation via reinforcement learning, graph-based associative structures, neuro-cognitive inspired forgetting mechanisms, and multi-agent collaborative memory systems. Our analysis reveals a paradigm shift from passive storage to active memory governance, where agents learn to compress, retrieve, update, and forget information dynamically. We critically evaluate methodological families, highlighting that while hierarchical and graph-based methods excel in structural reasoning, reinforcement learning-driven context management offers superior adaptability in dynamic environments. Furthermore, we discuss emerging challenges in safety, evaluation bottlenecks, and the trade-offs between computational overhead and reasoning fidelity, outlining a roadmap for future research in agentic memory systems.

1. Introduction

The evolution of Large Language Models (LLMs) from static text generators to autonomous agents capable of executing multi-step plans has fundamentally altered the landscape of artificial intelligence. Agents are increasingly tasked with long-horizon objectives that require sustaining coherence, tracking state, and integrating information across thousands of interaction turns or massive document corpora. Tasks such as autonomous software debugging [1], longitudinal scientific research [2], and personalized lifelong companionship [3] demand memory capabilities that exceed the limits of fixed context windows. Although recent models boast context lengths exceeding one million tokens, empirical evidence suggests that performance degrades significantly as context length increases, particularly for tasks requiring precise retrieval of distant information or complex causal reasoning [4], [5]. The phenomenon of “context rot,” where agents lose track of goals or succumb to noise injection, remains a critical bottleneck [5], [6].

Consequently, the field has witnessed an explosion of research dedicated to external and internal memory mechanisms designed to augment LLM agents. Early approaches relied on simple Retrieval-Augmented Generation (RAG) or sliding window summaries. However, contemporary research demonstrates that effective long-horizon agency requires more than static retrieval; it demands active memory management, structural organization, and adaptive forgetting. Recent works have introduced hierarchical memory systems that mimic human cognitive architectures, separating working, episodic, and semantic memory [7], [8], [9]. Others have framed memory operations as learnable actions within a reinforcement learning (RL) framework, allowing agents to optimize what to remember and when to forget based on task rewards [10], [11], [12]. Furthermore, graph-based structures have emerged as powerful substrates for encoding causal and temporal dependencies, enabling agents to traverse complex knowledge networks rather than relying on flat vector similarity [13], [14], [15].

This survey synthesizes the state-of-the-art in memory and long-context mechanisms for LLM agents, drawing upon a comprehensive corpus of 300 relevant academic papers published between 2021 and 2026. We move beyond a mere listing of techniques to provide a critical synthesis of methodological families. We examine how different architectures address the core challenges of long-horizon planning: information density, state persistence, interference mitigation, and computational efficiency. Specifically, we categorize existing literature into five primary directions: (1) Hierarchical and Multi-Layered Memory Architectures that structure information across temporal and abstraction scales; (2) Active Context Curation and Reinforcement Learning, where memory management is treated as a policy optimization problem; (3) Graph-Based and Associative Memory Systems that leverage structural relationships for reasoning; (4) Neuro-Cognitive and Biological Inspirations, including forgetting curves and consolidation mechanisms; and (5) Multi-Agent and Distributed Memory paradigms that decentralize memory responsibilities.

Our analysis highlights a converging consensus: the most effective agents treat memory not as a passive database but as an active, evolving component of the reasoning loop. Methods like HIAGENT [16] and MemAgent [17] demonstrate that summarizing trajectories into compact chunks preserves essential state information while reducing cognitive load. Similarly, approaches like Memex(RL) [18] and Agentic Memory [10] show that integrating memory operations directly into the agent’s action space via RL leads to robust performance in complex environments. However, significant challenges remain. Evaluation benchmarks often fail to capture the nuances of cross-session dependency [19], [20], and safety concerns regarding memory poisoning and hallucination propagation are increasingly prominent [21], [6].

By systematically reviewing these 300 studies, this paper aims to provide a definitive reference for researchers and practitioners. We identify key trends, such as the shift from heuristic compression to learned compression policies [22], the integration of causal reasoning into memory graphs [23], and the emergence of “memory as cognition” paradigms where retrieval is interleaved with reasoning [24]. We conclude by discussing open problems in scalability, safety, and the need for more realistic, dynamic evaluation frameworks that test agents in truly open-ended, lifelong scenarios.

2. Method

The methodological landscape of long-horizon LLM agent memory is diverse, spanning architectural innovations, learning algorithms, and structural representations. Based on the provided literature, we organize these approaches into five coherent families: Hierarchical and Multi-Layered Architectures, Active Context Curation via Reinforcement Learning, Graph-Based and Associative Memory Systems, Neuro-Cognitive and Biological Mechanisms, and Multi-Agent Collaborative Memory. Each family addresses specific limitations of flat context windows, offering distinct trade-offs between fidelity, efficiency, and reasoning capability.

2.1 Hierarchical and Multi-Layered Memory Architectures

A dominant theme in recent literature is the decomposition of memory into hierarchical layers, mirroring human cognitive models that distinguish between short-term working memory, episodic records, and abstract semantic knowledge. This stratification allows agents to manage vast amounts of information by operating at appropriate levels of abstraction, retrieving high-level summaries for planning and drilling down into details only when necessary.

2.1.1 Working, Episodic, and Semantic Decomposition The foundational concept of separating memory by temporal scope and abstraction level is central to many high-performing systems. The Multi-Layered Memory Framework (MLMF) explicitly decomposes dialogue history into working, episodic, and semantic layers [7]. In this architecture, working memory retains recent utterances via bounded encoding, while episodic memory accumulates session summaries through recursive blending with a decay parameter. Crucially, the semantic layer stores stable knowledge, and ablation studies reveal that removing this layer causes the largest drop in retention performance, underscoring the necessity of long-term abstraction [7]. Similarly, HiMem constructs a dual-layer system comprising Episode Memory for fine-grained context and Note Memory for stable knowledge, utilizing a topic-aware event-surprise segmentation to define cognitively consistent boundaries [9]. This hierarchical organization optimizes retrieval efficiency without sacrificing reasoning fidelity, outperforming flat baselines on multi-hop reasoning tasks [9].

Extending this concept, the Dual-Process Memory Architecture decouples immediate context (Episodic Buffer) from long-term knowledge (Neocortical Memory) using asynchronous consolidation [8]. This system triggers consolidation per message to extract facts and detect contradictions via temporal precedence rules, enabling sustained operation beyond context limits. The insight here is that episodic-semantic decomposition prevents memory interference by maintaining constant-size recent buffers while compressing older information [8]. Further refinement is seen in TiMem, which organizes conversations into a five-level Temporal Memory Tree (TMT) enforcing explicit temporal containment [25]. By distilling raw dialogue into factual segments and evolving patterns at different levels, TiMem enables progressive consolidation from fine-grained facts to abstract personas, achieving superior accuracy on temporal reasoning benchmarks [25].

2.1.2 Subgoal-Driven and Task-Oriented Chunking For agents engaged in complex planning, memory organization often aligns with task structure rather than purely temporal sequences. HIAGENT introduces a hierarchical working memory management system where the LLM partitions memory into subgoal-based chunks [16]. Upon completing a subgoal, the agent summarizes the trajectory into a condensed observation, obscuring detailed action-observation pairs to reduce cognitive load. This subgoal-driven chunking with automatic trajectory summarization has been shown to double success rates in long-horizon tasks compared to standard baselines [16]. Similarly, ReCAP executes recursive task decomposition within a shared context window, structurally re-injecting parent plans during backtracking to prevent context fragmentation [26]. This approach preserves global intent better than isolated subgoal prompting, demonstrating that maintaining a shared context tree is vital for hierarchical planning [26].

In the domain of software engineering, ContextWeaver constructs a Directed Acyclic Graph (DAG) where nodes represent Thought-Action-Observation triples linked by logical dependencies rather than recency [27]. This dependency-structured memory allows for incremental summarization and dynamic node pruning based on execution feedback, preserving critical context better than recency-based buffers [27]. The insight is that explicitly modeling logical dependencies between reasoning steps is superior to simple chronological storage for maintaining coherence in long-horizon coding tasks [27].

2.1.3 Hierarchical Indexing and Retrieval Efficient retrieval is as critical as storage structure. H-MEM organizes memory into four semantic layers (Domain, Category, Memory Trace, Episode) and embeds positional indices of sub-memories into parent vectors [28]. This enables index-based recursive routing, eliminating exhaustive vector similarity searches and achieving logarithmic time

complexity while preserving contextual relevance [28]. Similarly, MemForest decouples persistent state from derived artifacts, organizing memory into balanced temporal hierarchies (MemTrees) where leaves store time-local evidence and internal nodes summarize children [29]. This hierarchical temporal indexing allows for localized updates and parallel extraction, yielding 6x higher throughput than flat summarization systems [29].

The Hierarchical Long-Term Semantic Memory (HLTM) used in industrial hiring agents aligns memory topology with business schemas, creating a tree structure that enforces privacy via subtree isolation [30]. By generating multi-view representations (Facet, QA, Summary) for each node, HLTM enables bottom-up retrieval that respects organizational boundaries, demonstrating the practical utility of schema-aligned hierarchies in enterprise settings [30].

2.2 Active Context Curation and Reinforcement Learning

A significant paradigm shift in recent years is the treatment of memory management as an active decision-making process optimized via Reinforcement Learning (RL). Instead of relying on fixed heuristics for summarization or retrieval, these methods train agents to learn *when* to write, *what* to retrieve, and *how* to compress based on downstream task rewards.

2.2.1 Memory as Action and Policy Optimization The “Memory-as-Action” (MemAct) framework integrates context management directly into the policy action space via a Prune&Write operator [11]. The agent selects target interaction IDs for deletion and generates synthesized memory content simultaneously, optimizing this joint policy using Group Relative Policy Optimization (GRPO). This approach treats memory editing as an intrinsic policy action, enabling joint optimization of context efficiency and task accuracy [11]. Similarly, Agentic Memory (AgeMem) defines a hybrid action space containing language generation and explicit memory operations (ADD, UPDATE, DELETE, RETRIEVE, SUMMARY, FILTER) [10]. By employing a three-stage progressive RL with step-wise GRPO, AgeMem unifies Long-Term Memory (LTM) and Short-Term Memory (STM) management into a single policy, eliminating reliance on static schedules [10].

AtomMem further refines this by formulating memory management as a Partially Observable Markov Decision Process (POMDP) with atomic CRUD (Create, Read, Update, Delete) operations [31]. It decomposes static workflows into learnable actions, allowing agents to dynamically balance creation, retrieval, and deletion. The dual-retrieval mechanism combines a deterministic scratchpad for global state with selective vector search, maintaining performance even with 4x context extension where other methods degrade [31].

2.2.2 Learned Compression and Summarization Learning *how* to compress information is another critical area. MemAgent decomposes autoregressive generation into sequential read-write steps using a latent fixed-length memory variable [17]. It processes input text in chunks, overwriting previous memory states with updated summary tokens optimized via Multi-Conversation DAPO. This token-space memory overwrite strategy enables selective information retention without architectural changes, maintaining performance up to 512K context [17]. Similarly, Memo employs segmented context summarization with accumulation, generating learnable summary tokens at segment boundaries [32]. These accumulated tokens create residual-like gradient paths, stabilizing optimization over extended horizons better than static compression [32].

FoldAct addresses the instability of context folding in RL by introducing token-level separated loss computation [33]. It assigns independent gradients to summary and action tokens using binary

masks, preventing gradient dilution. Additionally, a full context consistency loss mitigates policy-dependent distribution shifts, ensuring that summary tokens receive distinct credit assignment signals [33]. ReSum employs a periodic context compression mechanism where a specialized summary tool condenses interaction history into compact reasoning states containing verified evidence and gaps [34]. Trained via ReSum-GRPO, this method converts growing histories into compact states, bypassing context constraints while preserving essential evidence [34].

2.2.3 Gating, Stopping, and Adaptive Control Active control also involves deciding *when* to stop retrieving or updating. InfMem executes a PRETHINK–RETRIEVE–WRITE protocol where a System-2 style controller monitors memory sufficiency to decide whether to STOP or RETRIEVE [35]. This state-dependent controller, aligned via verifier-based RL, mitigates context fragmentation better than reactive compression [35]. GRU-Mem augments recurrent memory agents with text-controlled Update and Exit Gates [36]. The Update Gate filters irrelevant chunks to prevent memory inflation, while the Exit Gate terminates the loop upon sufficient evidence collection, accelerating inference by 400% compared to vanilla agents [36].

Budget-Aware Context Management (BACM) formulates context management as a budget-constrained MDP, introducing a budget-conditioned state that exposes remaining capacity before observation loading [37]. It employs a commit-block aggregation policy to select compression levels dynamically, enabling adaptive trade-offs between fidelity and feasibility under tight token budgets [37].

2.3 Graph-Based and Associative Memory Systems

Flat vector stores often fail to capture the complex relational structures inherent in long-horizon tasks. Graph-based memory systems address this by explicitly modeling semantic, temporal, and causal relationships between memory units, enabling structured traversal and reasoning.

2.3.1 Semantic and Temporal Graphs GAM (Hierarchical Graph-based Agentic Memory) decouples memory into a local Event Progression Graph for rapid buffering and a global Topic Associative Network for stable retention [14]. A state-based mechanism triggers consolidation only upon semantic divergence, merging events into the global graph. This decoupling minimizes interference while preserving long-term consistency, outperforming flat baselines on temporal F1 scores [14]. SYNAPSE constructs a dual-layer directed graph partitioning episodic interaction logs and abstract semantic concepts, implementing spreading activation dynamics where query anchors inject energy propagating through temporal and causal edges [38]. This allows relevance to emerge from structural connectivity rather than pre-computed similarity, significantly improving adversarial robustness [38].

AriadneMem implements a two-phase pipeline where offline construction employs entropy-aware gating to filter noise and conflict-aware coarsening to merge duplicates, encoding state updates as directed temporal edges [39]. Online reasoning then executes algorithmic bridge discovery using approximate Steiner trees, transforming memory into an evolutionary graph that enables deterministic structural traversal [39]. Similarly, H-MEM (Hybrid Memory Structure) constructs an incremental temporal-semantic tree where leaf nodes store events and upper nodes store consolidated summaries, coupled with a parallel entity knowledge graph for multi-hop relations [40]. This hybridization enables simultaneous tracking of semantic consolidation over time and cross-fragment relational reasoning [40].

2.3.2 Causal and Dependency Graphs For tasks requiring rigorous reasoning, capturing causality is paramount. ActMem extracts atomic facts and constructs a memory knowledge graph integrating semantic edges and PMI-validated causal edges [23]. A counterfactual reasoning module generates alternative scenarios to bridge semantically distant queries, demonstrating that integrating causal graphs enables agents to uncover hidden historical constraints [23]. ContextWeaver also utilizes a DAG to link reasoning steps via logical dependency analysis, validating nodes through execution feedback to dynamically prune irrelevant history [27].

In the domain of web agents, LongSeeker augments the ReAct loop with a meta-action layer including operations like Skip, Compress, Rollback, and Delete to transform working context history [41]. While not strictly a graph, this elastic context orchestration relies on understanding the dependency structure of the search trajectory to apply these meta-operations effectively [41].

2.3.3 Spreading Activation and Associative Retrieval Retrieval in graph memory often leverages spreading activation. Field-Theoretic Memory (FTM) models memory as a scalar field on a 2D semantic manifold evolving via a reaction-diffusion equation [42]. Importance-weighted dynamics encode temporal structure and semantic proximity inherently through diffusion and decay terms, capturing associative spreading without explicit graph traversal [42]. HeLa-Mem models conversation as a dynamic graph where edge weights update via a Hebbian rule based on co-activation [43]. A Reflective Agent identifies high-degree hub nodes for distillation into stable semantic knowledge, enabling retrieval of latent associations beyond simple semantic similarity [43].

2.4 Neuro-Cognitive and Biological Mechanisms

Inspired by human cognition, several approaches incorporate mechanisms for forgetting, consolidation, and sleep-phase processing to manage memory capacity and prevent interference.

2.4.1 Forgetting Curves and Decay OBLIVION implements a self-adaptive read/write loop managed by a Decayer that gates retrieval via uncertainty estimation and Ebbinghaus-inspired retention scores [44]. By casting forgetting as accessibility decay rather than erasure, it enables dynamic filtering of high-utility information, preventing redundant access while retaining latent knowledge [44]. FadeMem employs a dual-layer hierarchy governed by adaptive exponential decay functions modulated by semantic relevance and frequency [45]. It utilizes LLM-guided conflict resolution to classify memory updates, mirroring biological consolidation to prevent cognitive overload [45].

The Adaptive Budgeted Forgetting Framework organizes interaction history into multi-layer components regulated by relevance-guided control, enforcing a fixed memory budget via constrained optimization [46]. This structured forgetting via relevance scoring preserves reasoning consistency better than uncontrolled accumulation [46]. Similarly, SAGE integrates the Ebbinghaus forgetting curve with linguistic optimization (MemorySyntax) to dynamically retain information, reducing cognitive load while maintaining essential context [47].

2.4.2 Consolidation and Sleep-Phase Processing Nemori operationalizes a Two-Step Alignment Principle via an LLM-based boundary detector for semantic segmentation and a narrative generator for episodic representation [48]. It implements a Predict-Calibrate cycle inspired by the free-energy principle, autonomously organizing raw streams into semantically coherent episodes [48]. The Biologically-Grounded Memory Architecture implements six cognitive mechanisms, including

sleep-phase consolidation that deduplicates events offline and interference-based forgetting [49]. This synthetic calibration methodology derives pipeline thresholds without benchmark exposure, driving quality improvement by removing redundancy [49].

RecMem buffers raw interactions in a subconscious vector store and triggers LLM-based consolidation only upon detecting semantic recurrence exceeding a threshold [50]. This recurrence-triggered consolidation defers expensive processing until patterns emerge, aligning with cognitive science principles that isolated experiences remain transient [50].

2.5 Multi-Agent and Distributed Memory Systems

To handle the complexity of long-horizon tasks, some systems distribute memory responsibilities across multiple specialized agents, decoupling planning, execution, and memory management.

2.5.1 Specialized Memory Agents E-mem decouples global planning from memory retention via a heterogeneous Master-Assistant tuple [51]. A central master agent orchestrates high-level synthesis while multiple Small Language Model (SLM) based assistant agents preserve immutable, uncompressed episodic contexts. This multi-pathway activation enables active re-experience and local reasoning, recovering details lost in compression [51]. MIRIX deploys eight specialized agents, including six Memory Managers for Core, Episodic, Semantic, Procedural, Resource, and Knowledge Vault components [52]. This segregation enables precise routing and reduces interference, active retrieval pipelines further enhancing performance on multimodal tasks [52].

AMA (Adaptive Memory via Multi-Agent Collaboration) decomposes the memory lifecycle into four agents: Constructor, Retriever, Validator, and Refiner [53]. Dynamic intent-based routing aligns retrieval precision with reasoning complexity, while logic-driven conflict detection triggers iterative refinement [53].

2.5.2 Collaborative and Parallel Architectures Chain-of-Agents (CoA) partitions long inputs into chunks processed by sequential worker agents that pass updated communication units to the next agent [54]. This sequential inter-agent communication preserves global semantic coherence better than parallel structures, outperforming vanilla long-context models on narrative QA [54]. EXTAGENTS partitions input into fixed-size chunks processed by parallel Seeking Agents, implementing global knowledge synchronization via a shared scratchpad [55]. This maximizes agent interaction bandwidth, resolving synchronization bottlenecks in ultra-long contexts [55].

Self-Manager architects a parallel single-agent loop where a main thread dynamically spawns isolated subthreads for subtask execution, encapsulated by Thread Control Blocks (TCBs) [56]. This prevents context contamination and allows concurrent exploration, narrowing the gap with proprietary models in deep research tasks [56].

2.6 Evaluation and Benchmarks

Robust evaluation is critical for advancing the field. Recent benchmarks have moved beyond simple QA to assess interdependent tasks, causal reasoning, and lifelong learning.

MEMORYARENA establishes a Memory-Agent-Environment loop where agents execute sequentially dependent subtasks across isolated sessions [19]. It reveals that state-of-the-art agents with saturated performance on long-context recall exhibit near-zero success rates in interdependent tasks, highlighting a critical gap in current capabilities [19]. AMA-Bench constructs a Causality Graph to

evaluate memory systems on latent inter-state causal dependencies, showing that memory system design is the primary performance bottleneck rather than backbone model capacity [57].

MINTEVAL constructs interference-heavy contexts via sequential updates across four domains to probe multi-target aggregation tasks [20]. It finds that memory construction and retrieval constitute the primary bottleneck, with systems exhibiting strong bias towards recent information [20]. UltraHorizon establishes partially observable environments requiring sustained hypothesis testing, revealing that LLM agents consistently underperform humans due to in-context locking and foundational capability gaps [58].

Other notable benchmarks include MemGUI-Bench for mobile GUI agents, which employs a progressive scrutiny pipeline to evaluate cross-session learning [59], and VitaBench 2.0, which evaluates personalized agents on temporally ordered task sequences with evolving preferences [60]. These benchmarks collectively underscore the need for evaluation frameworks that test agents in dynamic, multi-session, and interference-prone environments.

3. Challenges and Outlook

Despite significant progress, the field of memory and long-context mechanisms for LLM agents faces several persistent challenges and open problems.

3.1 Evaluation Bottlenecks and Benchmark Limitations

Current evaluation methodologies often fail to capture the complexities of real-world long-horizon agency. Many benchmarks rely on static contexts or synthetic users, failing to account for preference drift, conflicting signals, and the dynamic nature of real interactions [61], [60]. As noted in MEMORYARENA, agents that excel at recall fail catastrophically when tasks require maintaining causal dependencies across sessions [19]. Furthermore, the “illusion of diminishing returns” suggests that marginal single-step accuracy improvements compound hyperbolically, yet current metrics often overlook this cumulative effect [62]. There is a pressing need for benchmarks that simulate lifelong learning with continuous streams of data, evaluating not just recall but the agent’s ability to adapt, forget, and relearn over time [63].

3.2 Safety and Security Concerns

As agents accumulate memory over long horizons, they become vulnerable to novel security threats. Memory poisoning attacks, such as MINJA, demonstrate that malicious records can be injected via indication prompts, inducing agents to generate bridging steps that link benign queries to malicious targets [21]. The “Memory Curse” phenomenon reveals that expanded recall can erode cooperative intent in multi-agent social dilemmas, as agents become overly susceptible to negative evidence in their history [64]. Additionally, safety mechanisms like refusal rates exhibit unpredictable divergence as context length increases, with models suffering severe performance degradation beyond 100K tokens [6]. Developing robust defenses against memory injection and ensuring that forgetting mechanisms do not inadvertently delete safety-critical information remain critical research directions [65], [21].

3.3 Computational Efficiency and Scalability

While hierarchical and graph-based methods improve reasoning, they often incur significant computational overhead. Graph traversal, iterative consolidation, and multi-agent collaboration increase

latency and token consumption, limiting deployment in resource-constrained environments [13], [42], [53]. For instance, Field-Theoretic Memory incurs a 9.4x processing time increase, and many graph-based systems face scalability challenges with quadratic complexity [42], [66]. Efficient algorithms for approximate graph traversal, lightweight compression mechanisms, and hardware-software co-design (e.g., flattened systolic arrays for long-context inference) are essential to bridge the gap between theoretical performance and practical deployment [67], [29].

3.4 The Forgetting-Detail Trade-off

A fundamental tension exists between compression and fidelity. Aggressive summarization and forgetting mechanisms risk losing fine-grained details necessary for complex reasoning, while retaining full history leads to context overflow and noise [68], [69]. Studies show that forced abstraction can compound errors, leading to performance degradation below no-memory baselines [68]. Future research must focus on adaptive mechanisms that dynamically balance compression and retention based on task requirements, perhaps leveraging uncertainty estimation to identify critical information that must be preserved [35], [44].

3.5 Future Directions

Future research should explore the integration of world models with memory systems, allowing agents to simulate future trajectories and plan memory usage proactively [70], [71]. The development of “memory-native” architectures, where memory operations are intrinsic to the model rather than external add-ons, promises greater efficiency and coherence [72], [73]. Additionally, exploring the intersection of memory and multi-modal perception will be crucial for embodied agents operating in physical environments [3], [74]. Finally, fostering a deeper understanding of the theoretical limits of autoregressive reasoning and the role of memory in achieving universal computation will guide the design of next-generation agentic systems [75], [76].

4. Conclusion

The landscape of memory and long-context mechanisms for LLM agents has evolved rapidly from simple retrieval augmentation to sophisticated, active, and structured systems. This survey has synthesized 300 academic papers to highlight five key methodological families: hierarchical architectures, RL-driven context curation, graph-based associative systems, neuro-cognitive inspirations, and multi-agent collaborations. The consensus is clear: effective long-horizon agency requires treating memory as a dynamic, manageable resource rather than a static buffer. Hierarchical decomposition allows for efficient abstraction, RL enables adaptive control, graphs facilitate complex reasoning, biological inspirations offer robust forgetting mechanisms, and multi-agent systems distribute cognitive load.

However, significant challenges remain in evaluation, safety, and scalability. Current benchmarks often fail to reflect the dynamic, interdependent nature of real-world tasks, and security vulnerabilities like memory poisoning pose serious risks. As the field moves forward, the integration of world models, the development of memory-native architectures, and the creation of more rigorous, life-long evaluation frameworks will be paramount. By addressing these challenges, the community can unlock the full potential of LLM agents, enabling them to operate autonomously and reliably over indefinite horizons, ultimately bridging the gap between current capabilities and artificial general intelligence.

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